

**Diagnostic criteria for Transient Tic Disorder**

- A. Onset during childhood or early adolescence.
- B. Presence of recurrent, involuntary, repetitive, rapid, purposeless, motor movements (tics).
- C. Ability to suppress the movements voluntarily for minutes to hours.
- D. Variation in the intensity of the symptoms over weeks or months.
- E. Duration of at least one month but not more than one year.

**307.22 Chronic Motor Tic Disorder**

The essential features are recurrent, involuntary, repetitive, rapid movements (tics), usually involving no more than three muscle groups at any one time. The movements can be voluntarily suppressed for minutes to hours. The intensity of the symptoms is constant over weeks or months, and the duration is at least one year. Vocal tics occur infrequently. When present, they are not loud, intense or noticeable; frequently they are grunts or other noises caused by thoracic, abdominal, or diaphragmatic contractions.

**Age at onset.** The age at onset appears to be either in childhood or after age 40. When the onset is in adult life, the tic tends to be limited to a single muscle group.

**Course.** The course tends to be chronic.

**Complications, predisposing factors, sex ratio, familial pattern and prevalence.** No information.

**Differential diagnosis.** In **Transient Tic Disorder** the tics vary in intensity over time and the duration of the disturbance is always less than one year. In **Tourette's Disorder** the intensity of the tics varies over time, vocal tics are prominent and the motor movements are weaker and of briefer duration. See differential diagnosis of tics (p. 73) and Tourette's Disorder (p. 77).

**Diagnostic criteria for Chronic Motor Tic Disorder**

- A. Presence of recurrent, involuntary, repetitive, rapid, purposeless movements (tics) involving no more than three muscle groups at any one time.
- B. Unvarying intensity of the tics over weeks or months.
- C. Ability to suppress the movements voluntarily for minutes to hours.
- D. Duration of at least one year.

**307.23 Tourette's Disorder**

The essential features are recurrent, involuntary, repetitive, rapid movements (tics), including multiple vocal tics. The movements can be voluntarily suppressed for minutes to hours; and the intensity, frequency, and location of the symptoms vary over weeks or months.

The tics typically involve the head and, frequently, other parts of the body, such as the torso and upper and lower limbs. The vocal tics include various sounds such as clicks, grunts, yelps, barks, sniffs, and coughs, or words. Coprolalia, an irresistible urge to utter obscenities, is present in 60% of the cases. All of the symptoms are exacerbated by stress. They disappear, however, during sleep, and may become attenuated during some absorbing activities. Although the tics can be voluntarily suppressed, they eventually reappear.

**Associated features.** There may be other symptoms, such as echokinesis (imitation of the movements of someone who is being observed), palilalia (repetition of one's own last words or phrases), mental coprolalia (thinking about curse words), obsessive thoughts of doubting, and compulsive impulses to touch things or to perform complicated movements, such as squatting, deep knee bends, retracing steps, and twirling when walking.

Nonspecific EEG abnormalities, soft neurological signs, central nervous system psychological test abnormalities, hyperactivity or perceptual problems during infancy and childhood, or organic stigmata occur in about half the individuals with the disorder.

**Age at onset.** The disorder may appear as early as 2 years of age, and almost always appears before the age of 13.

**Course.** The disorder is usually lifelong, though brief periods of remission may occur. In some cases the disorder disappears before adulthood.

In approximately half the cases, the first symptom to appear is a single tic, most frequently an eye blink, less frequently a tic of another part of the face or the body. Initial symptoms also include tongue protrusion, squatting, sniffing, hopping, skipping, throat clearing, stuttering, uttering sounds or words, and coprolalia. Other cases begin with multiple symptoms, which may include any combination of the previously described tics and various noises such as barks, grunts, screams, yelps, or snorts.

**Complications.** In rare cases suicide may be a complication, because of despair over the disruptive effect of the disorder on social and occupational functioning.

**Predisposing factors.** There are no known predisposing factors. The disorder is unrelated to social class, or history of other mental disorders in the family or in the individual.

**Prevalence.** The estimated lifetime prevalence rate ranges from 0.1 to 0.5 per thousand.

**Sex ratio.** The disorder is three times more common in boys than in girls.

**Familial pattern.** Tics of various kinds are apparently more common among family members than in the general population.

**Differential diagnosis.** See differential diagnosis of tics, page 73. **Amphetamine Intoxication, cerebrovascular accidents, Lesch-Nyhan syndrome, Wilson's disease, Sydenham's and Huntington's chorea, multiple sclerosis, Schizophrenia, general paresis, and Organic Mental Disorders** may present with abnormal motor movements. These disorders can readily be distinguished from Tic Disorders because they have distinguishing symptoms, signs, clinical course, and physiological abnormalities as revealed by laboratory tests; and none of them have vocalizations similar to the clicks, grunts, yelps, barks, sniffs, coughs, and words of Tourette's Disorder.

#### **Diagnostic criteria for Tourette's Disorder**

A. Age at onset between 2 and 15 years.

B. Presence of recurrent, involuntary, repetitive, rapid, purposeless motor movements affecting multiple muscle groups.

C. Multiple vocal tics.

D. Ability to suppress movements voluntarily for minutes to hours.

E. Variations in the intensity of the symptoms over weeks or months.

F. Duration of more than one year.

#### **307.20 Atypical Tic Disorder**

This category is for the diagnosis of tics that cannot be adequately classified in any of the previous categories.

#### **307.30 Atypical Stereotyped Movement Disorder**

This category is for conditions such as head banging, rocking, repetitive hand movements consisting of quick, rhythmic, small hand rotations, or repetitive voluntary movements that typically involve the fingers or arms. These disorders are distinguishable from tics in that they consist of voluntary movements and are not spasmodic. Moreover, unlike individuals with a Tic Disorder, those with these conditions are not distressed by the symptoms and may even appear to derive enjoyment from the repetitive activities. Though bizarre posturing or movements may occur in adults, these conditions are found almost exclusively in children. They are especially prevalent among individuals with Mental Retardation or Pervasive Developmental Disorders and among children suffering from grossly inadequate social stimulation, but they may also occur in the absence of a concurrent mental disorder.

**OTHER DISORDERS WITH PHYSICAL MANIFESTATIONS**

This subclass of disorders includes categories in which the predominant disturbance is in a physical function: Stuttering (speech), Functional Enuresis and Functional Encopresis (urination and defecation), Sleepwalking Disorder and Sleep Terror Disorder (sleep). The inclusion of these categories in a classification of mental disorders is justified partly by tradition in that, formerly, psychological conflict was thought to play a central role in all of these disorders and it was thought that these conditions were almost always associated with other signs of psychopathology. Recently, however, many have come to question these assumptions, at least with regard to some of these categories. Further, there is evidence that most of the children with these disorders do not have associated mental disorder.

**307.00 Stuttering**

The essential features are frequent repetitions or prolongations of sounds, syllables, or words, or frequent, unusual hesitations and pauses that disrupt the rhythmic flow of speech. The extent of the disturbance varies from situation to situation and is most severe when there is special pressure to communicate, as during a job interview.

Speech may be very rapid or very slow, and there may be inappropriate inflection or lack of variation in pitch. In even the most severe cases, Stuttering is often absent during oral reading, singing, or talking to inanimate objects or to pets.

Stammering is a synonym for Stuttering.

**Associated features.** In moderate or severe cases there is vivid, fearful anticipation of stuttering with avoidance of particular sounds, words, or situations in which stuttering is anticipated. In addition, there may be eye blinks, tics, tremors of the lips or jaw, or jerking of the head.

**Age at onset.** Stuttering usually appears before the age of 12, with sharp peaks of onset between the ages of 2 to 3½ and 5 to 7 years.

**Course.** Typically the disturbance starts gradually with repetition of initial consonants, whole words that are usually the first words of a phrase, or long words. The child is generally not aware of the stuttering. As the disorder progresses, the repetitions become more frequent, and the stuttering occurs on the most important words of phrases. The child becomes aware of the speech difficulty, and in certain situations some words and sounds become more difficult.

The usual course is chronic, with periods of partial remission extending for weeks or months and exacerbations that occur most frequently when there is particular pressure to communicate. Between 50% and 80% of children with the disorder recover spontaneously; this is most likely in mild cases.

**Impairment.** Teasing and social ostracism by peers may result in impairment in peer relations. The child may have academic difficulties if he or she avoids speaking in class.

**Complications.** Limitation in occupational choice or advancement is the principal complication.

**Predisposing factors.** No information.

**Prevalence.** Approximately 1% of all children have a persistent problem with Stuttering that continues into adolescence.

**Sex ratio.** The disorder is about four times more common in males than in females.

**Familial pattern.** Stuttering is more common among family members than in the general population.

**Differential diagnosis.** *Spastic dysphonia*, a stuttering-like speech disorder, is distinguished from Stuttering by the presence of an abnormal pattern of breathing.

**Cluttering**, in which there is such a rapid rate of speech that fluency breaks down, may suggest Stuttering, but in cluttering the individual is usually unaware of the disturbance, whereas in Stuttering, after the initial phase, the individual is painfully aware of the disturbance.

#### Diagnostic criteria for Stuttering

Frequent repetitions or prolongations of sounds, syllables, or words or frequent, unusual hesitations and pauses that disrupt the rhythmic flow of speech.

### 307.60 Functional Enuresis

The essential feature is repeated involuntary voiding of urine during the day or at night, after an age at which continence is expected, that is not due to any physical disorder. The disorder is somewhat arbitrarily defined as involuntary voiding of urine at least twice a month for children between the ages of five and six and once a month for older children.

Functional Enuresis is often referred to as *primary* if it has not been preceded by a period of urinary continence for at least one year, and *secondary* if it has been preceded by a period of urinary continence for at least one year. Either of the above types may be *nocturnal* (most common), defined as the passage of urine during sleep time only, *diurnal*, defined as the passage of urine during waking hours, or *both* diurnal and nocturnal. There is no provision for coding these distinctions.

In most cases of nocturnal Functional Enuresis, the child awakens with no memory of a dream and no memory of having urinated because typically the disturbance occurs during the first third of the night, during non-REM sleep. However, in a few cases the voiding takes place during the rapid eye movement (REM) stage of sleep, and in such cases the child may recall a dream that involved the act of urinating.

**Associated features.** Very often the individual feels ashamed or embarrassed and may wish to avoid situations that might lead to embarrassment, such as camp or overnight visits to friends. Although the great majority of children with Functional Enuresis do not have a coexisting mental disorder, the prevalence of coexisting mental disorders is greater in individuals with Functional Enuresis than in the general population.

Functional Encopresis, Sleepwalking Disorder, and Sleep Terror Disorder may also be present.

**Course.** Most children with the disorder eventually become continent by adolescence, but in some cases the disorder continues into adulthood.

**Age at onset.** Primary Functional Enuresis by definition begins by age five. Most cases of secondary Functional Enuresis have their onset between the ages of five and eight.

**Impairment.** The amount of impairment is primarily a function of the effect on the individual's self-esteem, the degree of social ostracism by peers, and anger, punishment, and rejection from caretakers.

**Complications.** A complication is ascending urinary infection (particularly in girls).

**Predisposing factors.** These include delay in the development of the supporting musculature of the bladder, impaired ability of the bladder to adapt to urinary filling without changes in intravesical pressure with a lower bladder volume threshold for involuntary voiding, delayed or lax toilet training, and psychosocial stress (particularly hospitalization between the ages of two and four, entering school, and the birth of a sibling).

**Prevalence and sex ratio.** The prevalence of Functional Enuresis as defined here is: at age 5, 7% for boys, and 3% for girls; at age 10, 3% for boys, and 2% for girls; and at age 18, 1% for boys, and almost nonexistent for girls.

**Familial pattern.** Approximately 75% of all children with Functional Enuresis have a first-degree relative who has or has had the disorder. The concordance for the disorder is greater in monozygotic than in dizygotic twins.

**Differential diagnosis.** Organic causes of enuresis such as diabetes and seizure disorder should be ruled out by appropriate physical examination.

#### **Diagnostic criteria for Functional Enuresis**

A. Repeated involuntary voiding of urine by day or at night.

B. At least two such events per month for children between the ages of five and six, and at least one event per month for older children.

C. Not due to a physical disorder, such as diabetes or a seizure disorder.

### 307.70 Functional Encopresis

The essential feature is repeated voluntary or involuntary passage of feces of normal or near-normal consistency into places not appropriate for that purpose in the individual's own sociocultural setting, not due to any physical disorder.

Functional Encopresis is generally referred to as *primary* if it occurs after the child has reached the age of four and has not been preceded by fecal continence for at least one year and *secondary* if it has been preceded by a period of fecal continence for at least one year. There is no provision for recording the primary-secondary distinction. When the passage of feces in Functional Encopresis is involuntary rather than deliberate, it is often related to constipation, impaction, or retention with subsequent overflow. In such cases there often is soiling of clothes shortly after bathing because of reflex stimulation.

**Associated features.** Very often the individual feels ashamed or embarrassed and may wish to avoid situations that might lead to embarrassment, such as camp or overnight visits to friends. When the incontinence is clearly deliberate, antisocial and other psychopathological features are common. Smearing feces may be deliberate, and should be differentiated from smearing that takes place accidentally in the child's attempt to clear or hide feces passed involuntarily. Twenty-five percent of children with Functional Encopresis also have Functional Enuresis.

**Course.** Functional Encopresis rarely becomes chronic.

**Age at onset.** By definition primary Functional Encopresis begins by age four. Secondary Functional Encopresis usually begins between the ages of four and eight.

**Impairment.** The amount of impairment is primarily a function of the effect on the individual's self-esteem, the degree of social ostracism by peers, and anger, punishment, and rejection from caretakers.

**Complications.** None.

**Predisposing factors.** These include inadequate, inconsistent toilet training and psychosocial stress, such as entering school and the birth of a sibling.

**Prevalence.** It is estimated that approximately 1% of five-year-olds have the disorder. Primary Functional Encopresis apparently is more frequent in the lower socioeconomic classes.

**Sex ratio.** The disorder is more common in males than in females.

**Familial pattern.** No information.

**Differential diagnosis.** Functional Encopresis must be differentiated from structural organic causes of encopresis, such as aganglionic megacolon and anal fissure, which need to be ruled out by physical examination.

**Diagnostic criteria for Functional Encopresis**

- A. Repeated voluntary or involuntary passage of feces of normal or near-normal consistency into places not appropriate for that purpose in the individual's own sociocultural setting.
- B. At least one such event a month after the age of four.
- C. Not due to a physical disorder, such as aganglionic megacolon.

**307.46 Sleepwalking Disorder**

The essential features are repeated episodes of a sequence of complex behaviors that frequently, though not always, progress—without full consciousness or later memory of the episode—to leaving bed and walking about. The episode usually occurs between 30 and 200 minutes after onset of sleep (the interval of nonrapid eye movement [NREM] sleep that typically contains EEG delta activity, sleep stages 3 and 4) and lasts from a few minutes to about a half hour.

During a typical episode the individual sits up and carries out perseverative motor movements, such as picking at the blanket, and then performs semi-purposeful motor acts, which, in addition to walking, may include dressing, opening doors, eating, and going to the bathroom. The episode may terminate before the walking stage is reached.

During the episode the individual has a blank, staring face and is relatively unresponsive to the efforts of others to influence the sleepwalking or to communicate with him or her, and can be awakened only with great difficulty. During sleepwalking, coordination is poor; but the individual is able to see and walks around objects in his or her path. It is a myth that during sleepwalking the individual is careful and safe; in fact, he or she can stumble or lose balance and be injured by taking hazardous routes such as through windows or down fire escapes.

The walking behavior may terminate spontaneously by awakening followed by several minutes of disorientation. On the other hand, the individual may return to bed without ever reaching consciousness, or may lie down in another place to continue sleep, though mystified the next morning to find himself or herself there.

Upon awakening (either from the sleepwalking episode or the next morning), there is amnesia for the route traversed and what happened during the episode. Fragmentary dream images may be recalled, but not complete dream sequences.

Sleep EEG slow waves usually increase in amplitude in stage 4 sleep just preceding the episode; but EEG flattening, i.e., arousal, may occur before the episode. In the usual instance, as walking ensues, the high amplitude, slow wave pattern gives way to an admixture of NREM stages and lower amplitude EEG alpha activity.

Episodes are more likely if the individual is fatigued, has experienced a stress, or has taken a bedtime dose of a sedative or hypnotic substance.

**Associated features.** Frenzied behavior or aggression to persons or objects is infrequent. Sleepwalking may occur during sleepwalking, but articulation is poor; dialogue is rare.

Individuals with Sleepwalking Disorder have a higher than normal incidence of other episodic disorders associated with deep NREM sleep, such as nocturnal Functional Enuresis and Sleep Terror Disorder. There is no consistently associated psychopathology in children with this disorder. In contrast, adults with the disorder frequently do show evidence of other mental disorders, such as Personality Disorders.

**Age at onset.** Sleepwalking Disorder usually begins between ages 6 and 12. Adults who sleepwalk give a history of childhood episodes, then a period of remission until recurrence in the 20s and 30s.

**Course.** Sleepwalking usually lasts several years in children and adolescents, whether it occurs infrequently or nightly. The great majority of children or adolescents with the disorder are asymptomatic by their 20s; the disturbance tends to be more chronic in adults.

**Impairment.** Impairment is limited to avoidance of situations in which others might become aware of the disturbance, such as camp or overnight visits to friends.

**Complications.** Accidental injury during the episodes is the major complication.

**Predisposing factors.** Seizure disorders, CNS infections, and trauma are predisposing factors.

**Prevalence.** It is estimated that 1%–6% of children at some time have the disorder. As many as 15% of all children experience isolated episodes. Sleepwalking Disorder is rarer in adults.

**Sex ratio.** The disorder is apparently more common in males than in females.

**Familial pattern.** The disorder is more common among family members than in the general population. Family members of individuals with Sleepwalking Disorder tend to be deep sleepers.

**Differential diagnosis.** Psychomotor epileptic seizures may occur at night and produce episodes of perseverative behaviors similar to sleepwalking except that the individuals almost never return to their own beds. Also, during epileptic attacks there is total unreactivity to environmental stimuli, and perseverative motor movements such as swallowing and rubbing the hands are more common. Individuals with seizure disorders generally manifest such behaviors in the waking state as well, and the activity is associated with recordable seizure discharge. However, seizure disorders do not preclude coexisting Sleepwalking Disorder.

**Psychogenic Fugues** are distinguishable from Sleepwalking Disorder on several counts: Psychogenic Fugues are rare in children, typically begin in wakefulness, have a duration of hours or days, are not characterized by disturbances of consciousness, and are usually associated with other evidence of severe psychopathology.

**Sleep drunkenness** (prolonged transition to a clear consciousness after awakening) may resemble Sleepwalking Disorder except for the former's appearance in the morning and high frequency of aggressive behavior.

#### **Diagnostic criteria for Sleepwalking Disorder**

A. There are repeated episodes of arising from bed during sleep and walking about for several minutes to a half hour, usually occurring between 30 and 200 minutes after onset of sleep (the interval of sleep that typically contains EEG delta activity, sleep stages 3 and 4).

B. While sleepwalking, the individual has a blank, staring face; is relatively unresponsive to the efforts of others to influence the sleepwalking or to communicate with him or her; and can be wakened only with great difficulty.

C. Upon awakening (either from the sleeping episode or the next morning), the individual has amnesia for the route traversed and for what happened during the episode.

D. Within several minutes of awakening from the sleepwalking episode, there is no impairment of mental activity or behavior (although there may initially be a short period of confusion or disorientation).

E. There is no evidence that the episode occurred during REM sleep or that there is abnormal electrical brain activity during sleep.

#### **307.46 Sleep Terror Disorder**

The essential features are repeated episodes of abrupt awakening from sleep, usually beginning with a panicky scream. The episode usually occurs between 30 and 200 minutes after onset of sleep (the interval of nonrapid eye movement [NREM] sleep that typically contains EEG delta activity, sleep stages 3 and 4), and lasts 1 to 10 minutes. This condition has also been called *Pavor Nocturnus*.

During a typical episode, the individual sits up in bed with intense anxiety and displays agitated and perseverative motor movements (such as picking at the blanket), a frightened expression, dilated pupils, profuse perspiration, pilo-erection, rapid breathing, and quick pulse. An individual in this state is unresponsive to efforts of others to comfort him or her until the agitation and confusion subside. The individual may then recount having had a sense of terror and fragmentary dream images before arousal, but rarely a vivid and complete dream sequence. Morning amnesia for the entire episode is the rule. Episodes are

more likely if the individual is fatigued, has experienced a stress, or has taken a bedtime dose of a tricyclic antidepressant or neuroleptic.

Prior to a severe episode, the sleep EEG delta waves may be higher in amplitude than usual for the NREM phase of sleep, and breathing and heart-beat, slower. The onset of the episode is accompanied by a twofold to fourfold increase in heart rate, and the EEG quickly assumes an alpha pattern.

**Associated features.** There is no consistently associated psychopathology in children with this disorder. In contrast, adults with the disorder frequently do show evidence of other mental disorders, such as Generalized Anxiety Disorder.

**Age at onset.** Sleep Terror Disorder usually begins between ages 4 and 12. When the disorder begins in adulthood, it usually begins in the 20s or 30s; onset after 40 is rare.

**Course.** Episodes are extremely variable in frequency both within and among individuals, and usually occur at intervals of days or weeks, but may occur on consecutive nights. In children the disorder usually gradually disappears in early adolescence. When the disorder begins in adulthood, the course is often chronic.

**Impairment.** Impairment is limited to avoidance of situations in which others might become aware of the disturbance, such as camp or overnight visits to friends.

**Complications.** None.

**Predisposing factors.** No information.

**Prevalence.** It is estimated that 1% to 4% of children at some time have the disorder. A much greater proportion of children experience isolated episodes.

**Sex ratio.** The disorder is more common in males than in females.

**Familial pattern.** The disorder is apparently more common among family members than in the general population.

**Differential diagnosis.** REM sleep nightmares are distinguished from Sleep Terror Disorder by their appearance in the middle and latter thirds of the night, the milder anxiety experience, the absence of a panicky scream upon awakening, and the distinct recall of a detailed dream sequence in which a growing threat leads to awakening. Parents may misinterpret the fearfulness and fragmentary imagery reports of Sleep Terror Disorder as indicative of a REM sleep nightmare. Hypnagogic hallucinations may be associated with anxiety, but occur at onset of sleep and are vivid images at the transition from wakefulness to sleep. Epileptic seizures during sleep with postictal confusion may present a clinical picture similar to Sleep Terror Disorder, but the presence of epileptic seizures in the waking state or the presence of an abnormal sleep EEG rules out a diagnosis of Sleep Terror Disorder.

**Diagnostic criteria for Sleep Terror Disorder**

A. Repeated episodes of abrupt awakening (lasting 1-10 minutes) from sleep, usually occurring between 30 and 200 minutes after onset of sleep (the interval of sleep that typically contains EEG delta activity, sleep stages 3 and 4) and usually beginning with a panicky scream.

B. Intense anxiety during the episode and at least three of the following signs of autonomic arousal:

- (1) tachycardia
- (2) rapid breathing
- (3) dilated pupils
- (4) sweating
- (5) piloerection

C. Relative unresponsiveness to efforts of others to comfort the individual during the episode and, almost invariably, confusion, disorientation, and perseverative motor movements (e.g., picking at pillow).

D. No evidence that the episode occurred during REM sleep or of abnormal electrical brain activity during sleep.

**PERVASIVE DEVELOPMENTAL DISORDERS**

The disorders in this subclass are characterized by *distortions* in the development of multiple basic psychological functions that are involved in the development of social skills and language, such as attention, perception, reality testing, and motor movement.

In the past, children with these disorders have been described by many terms: Atypical Children, Symbiotic Psychotic Children, Childhood Schizophrenia, and others. Since these disorders apparently bear little relationship to the psychotic disorders of adult life, the term "psychosis" has not been used here in the name of this group of conditions. The term Pervasive Developmental Disorder has been selected because it describes most accurately the core clinical disturbance: many basic areas of psychological development are affected at the same time and to a severe degree.

Pervasive Developmental Disorders differ from the Specific Developmental Disorders in two basic ways. First, only a single specific function is affected in each Specific Developmental Disorder whereas in Pervasive Developmental Disorders multiple functions are always affected. Second, in Specific Developmental Disorders the children behave as if they are passing through an earlier normal developmental stage, because the disturbance is a *delay* in development, whereas children with Pervasive Developmental Disorders display severe qualitative abnormalities that are not normal for any stage of development, because the disturbance is a *distortion* in development.

The fifth digit should be used for all of the Pervasive Developmental Disorders to indicate whether the full syndrome is currently present (code=0) or

whether the full syndrome occurred previously, but now the individual has only residual symptoms of the disorder, such as blunted or inappropriate affect, social withdrawal, or eccentric behavior (code=1).

The ICD-9 category Disintegrative Psychosis is not included in this classification since the disorder apparently is a nonspecific Organic Brain Syndrome that consists of a Dementia plus other behavioral abnormalities, such as rapid loss of language and social skills. Such cases should be diagnosed as Dementia.

### **299.0x Infantile Autism**

The essential features are a lack of responsiveness to other people (autism), gross impairment in communicative skills, and bizarre responses to various aspects of the environment, all developing within the first 30 months of age. Infantile Autism may be associated with known organic conditions, such as maternal rubella or phenylketonuria. In such cases the behavioral syndrome Infantile Autism should be recorded on Axis I, and the physical disorder, on Axis III.

The relationship of this category to Schizophrenia is controversial. Some believe that Infantile Autism is the earliest form of Schizophrenia, whereas others believe that they are two distinct conditions. However, there is apparently no increased incidence of Schizophrenia in the families of children with Infantile Autism, which supports the hypothesis that the two disorders are unrelated.

The failure to develop interpersonal relationships is characterized by a lack of responsiveness to and a lack of interest in people, with a concomitant failure to develop normal attachment behavior. In infancy these deficiencies may be manifested by a failure to cuddle, by lack of eye contact and facial responsiveness, and by indifference or aversion to affection and physical contact. As a result, parents often suspect that the child is deaf. Adults may be treated as interchangeable, or the child may cling mechanically to a specific individual.

In early childhood there is invariably failure to develop cooperative play and friendships; but, as the children grow older, greater awareness of and attachment to parents and other familiar adults often develop. Some of the least handicapped may eventually reach a stage where they can become passively involved in other children's games or physical play such as running with other children. This apparent sociability is superficial, however, and can be a source of diagnostic confusion if mistaken for social relatedness when the diagnosis is made retrospectively.

Impairment in communication includes both verbal and nonverbal skills. Language may be totally absent. When it develops, it is often characterized by: immature grammatical structure, delayed or immediate echolalia, pronominal reversals (use of the pronoun "you" when "I" is the intended meaning), nominal aphasia (inability to name objects), inability to use abstract terms, metaphorical language (utterances whose usage is idiosyncratic and whose meaning is not clear), and abnormal speech melody, such as questionlike rises at ends of statements. Appropriate nonverbal communication, such as socially appropriate facial expressions and gestures, is often lacking.

Bizarre responses to the environment may take several forms. There may be resistance and even catastrophic reactions to minor changes in the environment,

e.g., the child may scream when his or her place at the dinner table is changed. There is often attachment to odd objects, e.g., the child insists on always carrying a string or rubber band. Ritualistic behavior may involve motor acts, such as hand clapping or repetitive peculiar hand movements, or insisting that fixed sequences of events precede going to bed. The fascination with movement may be exemplified by staring at fans, and the child may display inordinate interest in spinning objects. Music of all kinds may hold a special interest for the child. The child may be extremely interested in buttons, parts of the body, playing with water, or peculiar rote topics such as train schedules or historical dates. Tasks involving long-term memory, for example, recall of the exact words of songs heard years before, may be performed remarkably well.

**Associated features.** Mood may be labile; crying may be unexplained or inconsolable; there may be giggling or laughing without identifiable cause. There is often underresponsiveness or overresponsiveness to sensory stimuli, such as light, pain, or sound. Real dangers, such as moving vehicles and heights, may not be appreciated. Peculiar nervous habits, such as hair pulling or biting parts of the body, are sometimes present. Rocking or other rhythmic body movements also occur.

About 40% of children with the disorder have an IQ below 50; only 30% have an IQ of 70 or more. These children show extreme variability in intellectual functioning; they are often untestable on verbal tasks, and when testable, performance is worst on tasks demanding symbolic or abstract thought and sequential logic. However, tasks requiring manipulative or visual-spatial skills or immediate memory may be performed well.

**Age at onset.** By definition, the age at onset is always before 30 months. However, it may be difficult to establish age at onset retrospectively unless those who cared for the child during the early years are able to give accurate information about language development, sociability, and play. Parents of only children may be unaware of the problems until the child is observed with other children. The parents may then date the age at onset from that point, although a careful history will usually reveal that the abnormalities were present earlier.

**Course.** The disorder is chronic. Some of these children eventually are able to lead independent lives, with only minimal signs of the essential features of the disorder; but often the social awkwardness and ineptness persist (Residual State). Overall, one child in six makes an adequate social adjustment and is able to do some kind of regular work by adulthood; another one in six makes only a fair adjustment; and two-thirds remain severely handicapped and unable to lead independent lives. Factors related to long-term prognosis include IQ and development of language skills.

**Impairment.** The disorder is extremely incapacitating, and special educational facilities are almost always necessary.

**Complications.** The major complication is the development of epileptic sei-

zures secondary to an underlying physical disorder; about 25% or more of cases develop seizures in adolescence or early adult life. Most of the children with an IQ below 50 develop seizures, but only very few of those with normal intelligence do so.

**Prevalence.** The disorder is very rare (2-4 cases per 10,000). It is apparently more common in the upper socioeconomic classes, but the reason for this is not clear.

**Sex ratio.** The disorder is about three times more common in boys than in girls.

**Predisposing factors.** Maternal rubella (especially when associated with infantile deafness or blindness), phenylketonuria, encephalitis, meningitis, and tuberous sclerosis are among the predisposing factors. In the past, certain familial interpersonal factors were thought to predispose to the development of this syndrome, but recent studies do not support this view.

**Familial pattern.** The prevalence of Infantile Autism is 50 times as great in siblings of children with the disorder than in the general population.

**Differential diagnosis.** In **Mental Retardation** there are often behavioral abnormalities similar to those seen in Infantile Autism. However, the full syndrome of Infantile Autism is rarely present. When both disorders are present, both diagnoses should be made. In **Schizophrenia occurring in childhood** there are oddities of behavior; but typically there are hallucinations, delusions, and loosening of associations or incoherence, which are not present in Infantile Autism. In **Childhood Onset Pervasive Developmental Disorder**, the age at onset is later than in Infantile Autism and the full syndrome of Infantile Autism is not present. In children with **hearing impairments** there will be a history of responding consistently only to very loud sounds, whereas in Infantile Autism the response to sounds is inconsistent. An audiogram can rule out the possibility of hearing impairment. In **Developmental Language Disorder, Receptive Type**, the children generally make eye contact and will often try to communicate appropriately by means of gestures, whereas in Infantile Autism there is a pervasive lack of responsiveness.

#### **Diagnostic criteria for Infantile Autism**

- A. Onset before 30 months of age.
- B. Pervasive lack of responsiveness to other people (autism).
- C. Gross deficits in language development.
- D. If speech is present, peculiar speech patterns such as immediate and delayed echolalia, metaphorical language, pronominal reversal.

E. Bizarre responses to various aspects of the environment, e.g., resistance to change, peculiar interest in or attachments to animate or inanimate objects.

F. Absence of delusions, hallucinations, loosening of associations, and incoherence as in Schizophrenia.

### **299.00 Infantile Autism, Full Syndrome Present**

Currently meets the criteria for Infantile Autism.

### **299.01 Infantile Autism, Residual State**

#### **Diagnostic criteria for Infantile Autism, Residual State**

A. Once had an illness that met the criteria for Infantile Autism.

B. The current clinical picture no longer meets the full criteria for Infantile Autism, but signs of the illness have persisted to the present, such as oddities of communication and social awkwardness.

### **299.9x Childhood Onset Pervasive Developmental Disorder**

The essential features are a profound disturbance in social relations and multiple oddities of behavior, all developing after 30 months of age and before 12 years.

The disturbance in social relationships is gross and sustained, with such symptoms as lack of appropriate affective responsivity, inappropriate clinging, asociality, and a lack of peer relationships.

Oddities of behavior include: sudden excessive anxiety, constricted or inappropriate affect, resistance to change in the environment or insistence on doing things in the same manner every time, oddities of motor movement, speech abnormalities, hyper- or hypo-sensitivity to sensory stimuli, and self-mutilation. Examples of these behaviors are given in the diagnostic criteria.

**Associated features.** Frequently there are bizarre ideas and fantasies and preoccupation with morbid thoughts or interests. There may be pathological preoccupation with and attachment to objects, such as always carrying a string or rubber band. The disorder is particularly common in children with low IQs.

**Age at onset.** By definition the age at onset is after 30 months and before 12 years of age.

**Course.** The course is chronic. The long-term prognosis is probably better than that of Infantile Autism.

**Impairment.** The disorder is extremely incapacitating, and special educational facilities are regularly necessary.

**Complications.** The major complication is inability to function independently and the consequent continued need for supervision and financial support.

**Predisposing factors.** No information.

**Prevalence.** This is an extremely rare disorder.

**Sex ratio.** The disorder is far more common in boys than in girls.

**Familial pattern.** No information.

**Differential diagnosis.** In **Schizotypal Personality Disorder** there are also oddities of behavior and communication, but any disturbance in social relations is mild in comparison with the profound disturbance present in Childhood Onset Pervasive Developmental Disorder. Furthermore, symptoms such as disturbances in motor movement, inappropriate affect, and self-mutilation are not present in Schizotypal Personality Disorder.

For a discussion of the differential diagnosis with other conditions, see **Infantile Autism**, p. 89.

**Diagnostic criteria for Childhood Onset Pervasive Developmental Disorder**

A. Gross and sustained impairment in social relationships, e.g., lack of appropriate affective responsivity, inappropriate clinging, asociality, lack of empathy.

B. At least three of the following:

- (1) sudden excessive anxiety manifested by such symptoms as free-floating anxiety, catastrophic reactions to everyday occurrences, inability to be consoled when upset, unexplained panic attacks
- (2) constricted or inappropriate affect, including lack of appropriate fear reactions, unexplained rage reactions, and extreme mood lability
- (3) resistance to change in the environment (e.g., upset if dinner time is changed), or insistence on doing things in the same manner every time (e.g., putting on clothes always in the same order)
- (4) oddities of motor movement, such as peculiar posturing, peculiar hand or finger movements, or walking on tiptoe
- (5) abnormalities of speech, such as questionlike melody, monotonous voice
- (6) hyper- or hypo-sensitivity to sensory stimuli, e.g., hyperacusis
- (7) self-mutilation, e.g., biting or hitting self, head banging

C. Onset of the full syndrome after 30 months of age and before 12 years of age.

D. Absence of delusions, hallucinations, incoherence, or marked loosening of associations.

**299.90 Childhood Onset Pervasive Developmental Disorder, Full Syndrome Present**

Currently meets the criteria for Childhood Onset Pervasive Developmental Disorder.

**299.91 Childhood Onset Pervasive Developmental Disorder, Residual State****Diagnostic criteria for Childhood Onset Pervasive Developmental Disorder, Residual State**

A. Once had an illness that met the criteria for Childhood Onset Pervasive Developmental Disorder.

B. The current clinical picture no longer meets the full criteria for the disorder, but signs of the illness have persisted to the present, such as oddities of communication and social awkwardness.

**299.8x Atypical Pervasive Developmental Disorder**

This category should be used for children with distortions in the development of multiple basic psychological functions that are involved in the development of social skills and language and that cannot be classified as either Infantile Autism or Childhood Onset Pervasive Developmental Disorder.

**SPECIFIC DEVELOPMENTAL DISORDERS (AXIS II)**

This subclass is for disorders of specific areas of development not due to another disorder. For example, a delay in language development in an otherwise normal child would be classified as a Specific Developmental Disorder whereas a delay in language development in a child with Infantile Autism would be attributed to the Infantile Autism and therefore would not be classified as a Specific Developmental Disorder. Similarly, an individual with general delays in development would receive a diagnosis of Mental Retardation, not a Specific Developmental Disorder.

Each aspect of development noted here is related to biological maturation. However, there is no assumption regarding the primacy of biological etiological factors, and nonbiological factors are clearly involved in these disorders.

The inclusion of these categories in a classification of "mental disorders" is controversial, since many of the children with these disorders have no other signs of psychopathology, and the detection and treatment of the most common category, Developmental Reading Disorder, take place mainly within the educational system rather than the mental health system. Nevertheless, these conditions fall within the DSM-III concept of mental disorder (see p. 6); moreover, they are included in the mental disorders section of ICD-9 (see Appendix D).

Because Specific Developmental Disorders occur so frequently in conjunction with other disorders, they are coded on a separate axis (Axis II) to ensure that they are not overlooked. Thus, in the case of a child with Conduct Disorder and Developmental Reading Disorder, the Conduct Disorder will be coded on

Axis I, and the reading disorder on Axis II. A particular child may have more than one Specific Developmental Disorder; all should be diagnosed.

Although most of the clinical features seen in Specific Developmental Disorders represent functional levels that are normal for very young children (e.g., inability to do arithmetic), there is no implication that children with these disorders are simply at a lower end of a normal continuum and that they will "catch up" with time. In fact, children with these disorders frequently continue to show signs of the disturbance as adolescents or as adults; and the relevant diagnosis should be noted when an adult still has clinically significant signs of the disorder.

**Age at onset.** In these disorders onset is related to the age at which each area of functioning is expected to begin to develop. Thus, Developmental Language and Articulation Disorders may be first recognized within the first three years of life, when speech normally develops; and Developmental Reading and Arithmetic Disorders are first recognized in the first few years of school, when these skills normally develop.

**Course.** In most cases the disturbance is stable throughout childhood and adolescence, and many individuals continue to show some attenuated signs of the disturbance in adult life. More rarely, especially in mild cases, there is marked improvement in, or even disappearance of, all symptoms in time.

**Impairment.** Invariably there is some degree of impairment in academic functioning. The overall level of impairment is most marked when language or articulation is affected.

**Complications.** Academic failure, truancy, and antisocial behavior are among the commonly observed complications.

**Predisposing factors.** Factors that predispose one to most of these disorders are generally unknown. However, being a twin, being born prematurely or of an older mother, or sustaining a head injury apparently predispose to Developmental Reading Disorder.

**Sex ratio.** No information is available for Developmental Arithmetic Disorder. The other Specific Developmental Disorders are all about twice as common in males as in females.

### **315.00 Developmental Reading Disorder**

The essential feature is significant impairment in the development of reading skills not accounted for by chronological age, mental age, or inadequate schooling. In addition, in school, the child's performance on tasks requiring reading skills is significantly below his or her intellectual capacity. "Significant" impairment differs somewhat with age: a one-to-two-year discrepancy in reading skill for ages 8 to 13 is significant, but below that age, it is difficult to specify how great a discrepancy is significant.

This disorder has been referred to as "dyslexia." Faulty oral reading occurs, often characterized by omissions, additions, and distortions of words. Reading

is slow, and often there is reduced reading comprehension, although the ability to copy from written or printed texts is generally unaffected.

The diagnosis can be made only by individually administered IQ tests that contain verbal subtests and that yield a level of full-scale IQ, plus a variety of academic achievement tests that contain reading subtests.

**Associated features.** In spelling to dictation there may be numerous and bizarre errors that are not explainable by phonetics or by simple reversal of letters (such as b—d). Other common associated features include subtle language difficulties, such as impaired sound discrimination and difficulties with sequencing words properly, and behavioral problems, such as those associated with Attention Deficit Disorder and Conduct Disorder. “Soft” neurological signs, such as finger agnosia, may be found, particularly in younger children.

**Prevalence.** The disorder is apparently common.

**Familial pattern.** Reading difficulty and speech and language problems are more common in family members than in the general population.

**Differential diagnosis.** In **Mental Retardation**, reading difficulty is due to a general impairment in intellectual functioning. However, in some cases of Mild Mental Retardation, the reading level is significantly below the expected level, given the individual’s schooling and level of retardation. In such cases the additional diagnosis of Developmental Reading Disorder should be made, since treatment of the reading difficulties can greatly increase occupational potential.

**Inadequate schooling** can result in poor performance on standardized reading tests. In such cases, however, other children in the school will generally have similar difficulty. **Impaired vision or hearing** may affect reading ability, and can be ruled out with screening tests.

#### **Diagnostic criteria for Developmental Reading Disorder**

Performance on standardized, individually administered tests of reading skill is significantly below the expected level, given the individual’s schooling, chronological age, and mental age (as determined by an individually administered IQ test). In addition, in school, the child’s performance on tasks requiring reading skills is significantly below his or her intellectual capacity.

#### **315.10 Developmental Arithmetic Disorder**

The essential feature is significant impairment in the development of arithmetic skills not accounted for by chronological age, mental age, or inadequate schooling. In addition, in school, the child’s performance on tasks requiring arithmetic skills is significantly below his or her intellectual capacity. The diagnosis can be made only by individually administered IQ tests that yield a level of full-scale IQ, plus a variety of academic achievement tests containing arithmetic subtests.

**Associated features.** Other academic problems, including reading and spell-

ing difficulty, are often present; but generally the degree of these deficits is not as pronounced.

**Prevalence.** The disorder is apparently not common.

**Familial pattern.** No information.

**Differential diagnosis.** See Developmental Reading Disorder (p. 94).

#### **Diagnostic criteria for Developmental Arithmetic Disorder**

Performance on standardized, individually administered tests of arithmetic achievement is significantly below expected level, given the individual's schooling, chronological age, and mental age (as determined by an individually administered IQ test). In addition, in school, the child's performance on tasks requiring arithmetic skills is significantly below his or her intellectual capacity.

#### **315.31 Developmental Language Disorder**

There are three major types of language disorder: (1) failure to acquire any language, (2) acquired language disability, (3) delayed language acquisition (Developmental Language Disorder). Failure to acquire any language is rare and virtually always is a result of profound Mental Retardation. Acquired language disabilities are usually the result of trauma or neurological disorder. Developmental Language Disorder, the most common type of language disorder, involves difficulty in comprehending oral language (Receptive Type) or in expressing verbal language (Expressive Type). (Although these two subtypes are described separately, no digit is available for indicating them separately.)

These conditions have each been referred to as Developmental Aphasia, but this is technically not correct, since aphasia means loss of language that has already been acquired.

#### **Developmental language disorder: expressive type**

The essential feature is failure to develop vocal expression (encoding) of language while understanding or decoding skills remain relatively intact. Mothers will report that their children seem to understand language, but "can't get the words out." Articulation is generally immature, with the more difficult sounds, e.g., th, r, s, z, y, l, being omitted or other sounds substituted for them. The child's vocabulary is severely restricted; and even by four years of age, the child usually is unable to generate more than short phrases. Old words appear to be forgotten when new ones are learned. The child's use of various grammatical structures is considerably below age level.

**Associated features.** There may be slight lags in achieving developmental milestones. Learning may be impaired, particularly in tasks involving perceptual skills or skills in recognizing and reproducing symbols in the proper sequence.

**Prevalence.** Preliminary studies indicate that 1 in every 1,000 children may have this disorder.

**Familial pattern.** Developmental Articulation Disorder and other Specific Developmental Disorders are more common in family members of individuals with Developmental Language Disorder, Expressive Type, than in the general population.

**Differential diagnosis.** In **Developmental Language Disorder, Receptive Type**, comprehension of language is impaired, whereas in **Developmental Language Disorder, Expressive Type**, language comprehension is within normal limits. In **Developmental Articulation Disorder**, expressive language (vocabulary and grammar) is within normal limits, whereas in **Developmental Language Disorder, Expressive Type**, expressive language is impaired.

In **Mental Retardation**, there is general impairment in intellectual functioning, whereas in **Developmental Language Disorder, Expressive Type**, nonverbal intelligence is within normal limits. With a **hearing impairment**, a child does not have a normal audiogram and does not respond normally to sounds, whereas in **Developmental Language Disorder, Expressive Type**, audiogram and response to sounds are normal. In **Infantile Autism** and in **Childhood Onset Pervasive Developmental Disorder**, there is no "inner language," imaginary play, use of gestures, or warm social relationships, whereas these are all present in children with **Developmental Language Disorder, Expressive Type**.

In **acquired aphasia**, normal language is followed by onset of language disorder that may be associated with head trauma, seizures, or EEG abnormalities, whereas in **Developmental Language Disorder, Expressive Type**, normal language has not developed.

#### **Diagnostic criteria for Developmental Language Disorder, Expressive Type**

A. Failure to develop vocal expression (encoding) of language despite relatively intact comprehension of language.

B. Presence of inner language (the presence of age-appropriate concepts, such as understanding the purpose and use of a particular household object).

C. Not due to Mental Retardation, Childhood Onset Pervasive Developmental Disorder, hearing impairment, or trauma.

#### **Developmental language disorder: receptive type**

The essential feature is failure to develop comprehension (decoding) and vocal expression (encoding) of language. Deficits occur in sensory perception (recognition of auditory symbols [sounds] or visual symbols [pictures]), integration (ability to relate or manipulate auditory or visual symbols [e.g., recognizing that

a shoe and a sock are somehow related]), storage recall (ability to reproduce a sequence of auditory or visual stimuli some time after it has been presented), and "sequencing" (ability to recognize or reproduce sequences of symbols).

**Associated features.** A partial hearing defect for pure tones, resistance to auditory arousal, and inability to localize sound sources are common in this disorder. There often is a mild delay in the development of motor skills. Reading and spelling difficulties are invariably present. EEG abnormalities, usually bilateral, have been reported in some cases.

**Prevalence.** The rate of occurrence may be as high as 1 in 2,000.

**Familial pattern.** Seizures and all Specific Developmental Disorders, especially Developmental Reading Disorder, are apparently more common among family members of individuals with Developmental Language Disorder, Receptive Type than in the general population.

**Differential diagnosis.** In **Developmental Language Disorder, Expressive Type**, the child's comprehension of language is within normal limits for age level, whereas in **Developmental Language Disorder, Receptive Type**, language comprehension is below age level. In **Developmental Articulation Disorder**, both expression and comprehension of language are within normal limits for the child's age level, whereas in **Developmental Language Disorder, Receptive Type**, these are both impaired. In **Mental Retardation** there is general impairment in intellectual functioning, whereas in **Developmental Language Disorder, Receptive Type**, the child has at least normal nonverbal IQ. In **hearing impairment**, the child will have a history of responding only to very loud sounds, whereas in **Developmental Language Disorder, Receptive Type**, there may be a history of variable and inconsistent responses to sounds, the child often responding more to environmental sounds than to speech sounds. Abnormal audiometric test results occur in both hearing impairment and in **Developmental Language Disorder, Receptive Type**. In **Infantile Autism** and in **Childhood Onset Pervasive Developmental Disorder**, no efforts are made to communicate or watch faces, whereas in **Developmental Language Disorder, Receptive Type**, the children will make eye contact and will often try to communicate through gestures. In **Infantile Autism**, **Childhood Onset Pervasive Developmental Disorder**, and **Developmental Language Disorder, Receptive Type**, there are short auditory memory spans, auditory discrimination problems, and abnormalities of pitch and intonation.

In **acquired aphasia**, normal language is followed by a sudden onset of language problems, whereas in **Developmental Language Disorder, Receptive Type**, there is failure to develop language.

#### **Diagnostic criteria for Developmental Language Disorder, Receptive Type**

A. Failure to develop comprehension (decoding) and vocal expression (encoding) of language.

B. Not due to hearing impairment, trauma, Mental Retardation, or Childhood Onset Pervasive Developmental Disorder.

### **315.39 Developmental Articulation Disorder**

The essential feature is failure to develop consistent articulations of the later-acquired speech sounds, such as r, sh, th, f, z, l, or ch. Omissions occur or substitutions are made for these sounds, giving the impression of "baby talk." Vocabulary and grammatical structures are within age norms. This disorder encompasses a range from the misarticulation of one sound (e.g., l or n as in "lalling") to mispronouncing several sounds (e.g., s, z, sh, ch as in "lissing").

**Associated features.** Mild reading difficulties may be present.

**Prevalence and sex ratio.** The disorder is present in about 6% of male and 3% of female school-aged children.

**Familial pattern.** The disorder is more common among family members than in the general population.

**Differential diagnosis.** In **Developmental Language Disorder, Mental Retardation, Infantile Autism, and Childhood Onset Pervasive Developmental Disorder**, language development is impaired, whereas in **Developmental Articulation Disorder**, language development is normal. In **hearing impairment**, audiometric testing will reveal an abnormality, whereas in **Developmental Articulation Disorder**, hearing will be normal. In **dysarthria** (abnormal articulation due to disorders of the oral speech mechanism or to neurological abnormalities), there may be problems with chewing or sucking, drooling, the rate of speech may be slowed down, and vowel sounds may be affected. None of these are present in **Developmental Articulation Disorder**.

#### **Diagnostic criteria for Developmental Articulation Disorder**

A. Failure to develop consistent articulations of the later-acquired speech sounds, such as r, sh, th, f, z, l, or ch.

B. Not due to Developmental Language Disorder, Mental Retardation, Childhood Onset Pervasive Developmental Disorder, or physical disorders.

### **315.50 Mixed Specific Developmental Disorder**

This category should be used when there is more than one Specific Developmental Disorder, but none is predominant. It is common for a delay in the development of one skill (e.g., reading, arithmetic, or language) to be associated with delays in other skills. The Mixed Specific Developmental Disorder category should be used when the mixture of delayed skills is such that all skills are impaired to relatively the same degree. When the skills are impaired to varying

degrees, multiple diagnoses should be recorded, the skill most seriously impaired being recorded first.

**315.90 Atypical Specific Developmental Disorder**

This is a residual category for use when there is a Specific Developmental Disorder not covered by any of the previous specific categories.

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# Organic Mental Disorders\*

In DSM-III a distinction is made between organic *brain syndromes* and organic *mental disorders*. "Organic brain syndrome" is used to refer to a constellation of psychological or behavioral signs and symptoms without reference to etiology (e.g., Delirium, Dementia); "organic mental disorder" designates a particular organic brain syndrome in which the etiology is known or presumed (e.g., Alcohol Withdrawal Delirium, Multi-infarct Dementia).

By tradition, disorders that are related either to aging of the brain or to the ingestion of a substance are classified as mental disorders. For this reason Primary Degenerative Dementia, Multi-infarct Dementia, and the various Substance-induced Organic Mental Disorders are included in Section 1 of this chapter, beginning on page 124. Organic Mental Disorders whose etiology or pathophysiological process is either a diagnosis not included in the mental disorders section of ICD-9-CM or is unknown are in Section 2, page 162.

The essential feature of all these disorders is a psychological or behavioral abnormality associated with transient or permanent dysfunction of the brain. Organic Mental Disorders are diagnosed (*a*) by recognizing the presence of one of the Organic Brain Syndromes, as described below, and (*b*) by demonstrating, by means of the history, physical examination, or laboratory tests, the presence of a specific organic factor judged to be etiologically related to the abnormal mental state. Under certain circumstances, however, a reasonable inference of an organic factor can be made from the clinical features alone, in which case only step (*a*) may be necessary for diagnosis.

Organic Mental Disorders are a heterogeneous group; therefore, no single description can characterize them all. The differences in clinical presentation reflect differences in the localization, mode of onset, progression, duration, and nature of the underlying pathophysiological process.

Differentiation of Organic Mental Disorders as a separate class does not imply that nonorganic ("functional") mental disorders are somehow independent of brain processes. On the contrary, it is assumed that all psychological processes, normal and abnormal, depend on brain function. Limitations in our knowledge, however, sometimes make it impossible to determine whether a given mental disorder in a given individual should be considered an organic mental disorder (because it is due to brain dysfunction of *known* organic etiology) or whether it should be diagnosed as other than an Organic Mental Disorder (because it is more adequately accounted for as a response to psychological or social

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\* This chapter begins with a general description of Organic Mental Disorders as a broad category. It then describes ten specific Organic Brain Syndromes. Finally, each of the various Organic Mental Disorders is described.

factors [as in Adjustment Disorder] or because the presence of a specific organic factor has not been established [as in Schizophrenia]).

The organic factor responsible for an Organic Mental Disorder may be a primary disease of the brain or a systemic illness that secondarily affects the brain. It may also be a substance or toxic agent that is either currently disturbing brain function or has left some long-lasting effect. Withdrawal of a substance on which an individual has become physiologically dependent is another cause of Organic Mental Disorder.

The most common Organic Brain Syndromes are Delirium, Dementia, Intoxication, and Withdrawal. These syndromes display great variability among individuals and in the same individual over time. More than one Organic Brain Syndrome may be present in an individual simultaneously (e.g., Delirium superimposed upon Dementia), and one Organic Brain Syndrome may succeed another (e.g., thiamine-deficiency Delirium [Wernicke's encephalopathy] followed by Alcohol Amnestic Disorder [Korsakoff's disease]).

**Associated features.** A wide variety of different emotional, motivational, and behavioral abnormalities are associated with Organic Mental Disorders. It is often impossible to decide whether the symptoms are the direct result of damage to the brain or are a reaction to the cognitive deficits and other psychological changes that constitute the essential features of these disorders.

Severe emotional disturbances may accompany cognitive impairment in a person who views it as a loss, a serious threat to self-esteem, or both. Anxiety, depression, irritability, and shame of varying degrees of intensity may be present.

Compulsive individuals tend to be particularly intolerant of and disturbed by their reduced cognitive capacity or by perceptual abnormalities, such as hallucinations. They tend to react with a fear of loss of control. There may also be severe depression leading to suicidal attempts. Other individuals cope with memory and other cognitive deficits by marked orderliness, which helps them maintain some degree of control. Such people insist on keeping things in exactly the same places, keeping detailed notes and diaries to counteract forgetfulness, and avoiding situations in which their deficits could be exposed. Some people resort to circumstantiality and confabulation in an effort to conceal gaps in memory.

Paranoid attitudes and actual delusions may be exhibited by habitually suspicious individuals who feel threatened by their cognitive impairment. They may accuse others of maliciously misplacing or stealing their possessions. Irritability and outbursts of temper, sometimes with physical aggression, may occur. Some individuals are euphoric; others, apathetic.

Decreased control over sexual, aggressive, and acquisitive impulses may accompany cognitive impairment. Social judgment may be impaired and result in inappropriate behavior that provokes retaliation. Inappropriate sexual advances, exhibitionistic acts, stealing, ravenous eating, and other manifestations of faulty impulse control may be exhibited.

These associated features reflect the individual's personality, educational level, and interpersonal relations as well as the type and severity of the cogni-

tive impairment or abnormality. They are not necessarily correlated with the degree of cognitive impairment: a person with a relatively mild cognitive impairment may display conspicuous emotional and behavioral disturbance. Severe and widespread brain dysfunction, such as follows extensive neuronal destruction or metabolic dysfunction, may produce apathy, lethargy, incontinence of urine or feces, diminished psychomotor activity, somnolence, or blunting of affect.

Emotional and behavioral disturbances may result in social isolation, by withdrawal or ostracism; and this, in turn, tends to aggravate the cognitive disability.

**Age at onset.** Organic Mental Disorders may occur at any age. Delirium is most apt to occur at the extremes of the life cycle, while Dementia is most common in the elderly.

**Course.** Since Organic Mental Disorders encompass such a wide range of psychopathological syndromes and organic etiologies, no single course characterizes them all. Onset may be sudden, as in the case of Delirium associated with an acute infection or Dementia resulting from major head trauma, or it may be insidious, as in Primary Degenerative Dementia or the personality disturbance sometimes associated with temporal lobe epilepsy.

The course is also extremely variable. It may be steadily or irregularly progressive, episodic, static, or rapidly or gradually resolving. A major factor in determining the course is the nature of the underlying pathology. Metabolic disorders, substance intoxications and withdrawals, and systemic illnesses tend to cause temporary brain dysfunction, and may be followed by full recovery. Pathological processes causing structural damage to the brain are more likely to cause permanent residual impairment.

## ORGANIC BRAIN SYNDROMES

The Organic Brain Syndromes can be grouped into six categories:

1. **Delirium and Dementia**, in which cognitive impairment is relatively global;
2. **Amnestic Syndrome and Organic Hallucinosi\*s**, in which relatively selective areas of cognition are impaired;
3. **Organic Delusional Syndrome\*** and **Organic Affective Syndrome\***, which have features resembling Schizophrenic or Affective Disorders;
4. **Organic Personality Syndrome\***, in which the personality is affected;
5. **Intoxication and Withdrawal**, in which the disorder is associated with ingestion or reduction in use of a substance and does not meet the criteria for

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\* When the etiological agent is a substance, the term "Organic" is dropped and the name of the substance is substituted. For example, Organic Hallucinosi\*s due to alcohol is called Alcohol Hallucinosi\*s.

any of the previous syndromes (Strictly speaking, these two Organic Brain Syndromes are etiologically rather than descriptively defined.);

6. **Atypical or Mixed Organic Brain Syndrome**, which constitutes a residual category for any other Organic Brain Syndrome not classifiable as one of the previous syndromes.

(This manual does not divide the Organic Brain Syndromes into psychotic and nonpsychotic or acute and chronic [irreversible] forms, as have other classifications. Whereas these distinctions were made on the basis of severity, mode of onset, and presumptions concerning prognosis, the present classification is based on clinical symptoms alone. Delirium may, however, be said to be roughly equivalent to the DSM-I concept of acute brain syndrome, and Dementia, to that of chronic brain syndrome.)

### **Delirium**

The essential feature is a clouded state of consciousness, that is, a reduction in the clarity of awareness of the environment. This is manifested by difficulty in sustaining attention to both external and internal stimuli, sensory misperception, and a disordered stream of thought. In addition, disturbances of sleep-wakefulness and psychomotor activity are present. The onset is relatively rapid, and the course typically fluctuates. The total duration is usually brief.

In DSM-I this syndrome was called "acute brain syndrome." It has also been termed "acute exogenous reaction type," "acute confusional state," "toxic psychosis," and "metabolic encephalopathy." Furthermore, some reserve the term "delirium" for a particular, agitated variety of confusional state with vivid visual hallucinations. In this manual, however, Delirium is intended to include the broad spectrum of clinical states having in common the essential features described above.

In Delirium there is difficulty shifting, focusing, and sustaining attention. The individual is easily distracted by irrelevant stimuli. It may be difficult, or impossible, to engage him or her in conversation because attention wanders.

Perceptual disturbances are common and result in various misinterpretations, illusions, and hallucinations. For example, the banging of a door may be mistaken for a pistol shot (misinterpretation); the folds of the bedclothes may appear to be animate objects (illusion); or the individual may "see" a group of people hovering over the bed when no one is actually there (hallucination). Although sensory misperceptions and hallucinations are most commonly visual, they may occur in other sensory modalities as well. Misperceptions range from simple and uniform to highly complex. There are often both a delusional conviction of the reality of hallucinations and an emotional and behavioral response in keeping with their content.

The individual with Delirium cannot maintain a coherent stream of thought. Thinking loses its usual clarity and direction toward a goal; it appears fragmented and disjointed. In mild Delirium, this may be manifested by acceleration or slowing of thought; in severe Delirium, thinking may be totally disorganized. This disturbance is reflected in speech that, in some cases, is limited and sparse, and in others, pressured and incoherent, with unpredictable switching

from subject to subject. It is also reflected in defective reasoning and impaired goal-directed behavior. Perseveration of speech and behavior may appear. Although other cognitive disturbances, particularly disorientation and memory impairment, are also present in Delirium, the person may be so inattentive and incoherent that these mental functions cannot be meaningfully assessed.

The sleep-wakefulness cycle is almost invariably disturbed. This frequently involves some depression in the level of consciousness, ranging from simple drowsiness, through increasing stages of torpor, to stupor or semicoma. On the other hand, some individuals with Delirium are hypervigilant and have difficulty in falling asleep. Fluctuations from hypersomnolence to insomnia and reversals of the customary sleep-waking cycle may also be present. Vivid dreams and nightmares are common, and may merge with hallucinations.

Psychomotor activity is also disturbed. Many individuals are restless and hyperactive. Groping or picking at the bedclothes, attempting to get out of bed, striking out at nonexistent objects, and sudden changes of position are manifestations of increased psychomotor activity. On the other hand, there may be decreased psychomotor activity, with sluggishness and even certain features resembling catatonic stupor. Psychomotor activity often shifts abruptly from one of these extremes to another.

**Associated features.** Emotional disturbances are very common and quite variable. They include anxiety, fear, depression, irritability, anger, euphoria, and apathy. Some individuals maintain the same emotional tone throughout the course of the Delirium whereas others experience rapid and unpredictable changes from one emotional state to another. Fear is very commonly experienced, sometimes as the result of threatening hallucinations or poorly systematized delusions. If fear is marked, the individual may attempt to flee his or her surroundings without regard to possible injury, or may attack those who are falsely viewed as threatening. Severely depressed feelings may also lead to self-destructive acts. Crying, calls for help, cursing, muttering, moaning, and other vocal productions, particularly prominent at night, are further manifestations of the disturbed emotional state of individuals with Delirium.

Neurological signs are comparatively uncommon in Delirium. An important exception, however, is the presence of abnormal movements. Various forms of tremor are frequently present. Asterixis, a peculiar flapping movement of the hyperextended hands, was originally described in hepatic encephalopathy, but may be found in other delirious states as well. Autonomic signs (tachycardia, sweating, flushed face, dilated pupils, and elevated blood pressure) commonly occur.

Other disorders of higher cortical function in Delirium include dysnomia (inability to name objects) and dysgraphia (impaired ability to write).

**Age at onset.** Delirium can occur at any age, but is especially common in children and after the age of 60.

**Course.** Delirium usually develops over a short period of time. Sometimes it begins quite abruptly, e.g., after a head injury or following a seizure. At other

times it is preceded, for hours or days, by certain prodromal symptoms. These include restlessness, difficulty in thinking clearly, hypersensitivity to auditory and visual stimuli, nocturnal insomnia, daytime hypersomnolence, vivid dreams, and nightmares. The slower evolution is more likely if systemic illness or metabolic imbalance underlies the Delirium.

Fluctuation in symptoms is one of the hallmarks of Delirium. Typically, the individual is worse during sleepless nights or in the dark. So-called "lucid intervals," periods during which he or she is more attentive and coherent, may occur at any time, but are most common in the morning. These fluctuations help distinguish Delirium from other brain syndromes.

The duration of an episode of Delirium is usually brief, about one week; it is rare for Delirium to persist for more than a month. If the underlying disorder is promptly corrected or is self-limited, recovery from Delirium may be complete. On the other hand, if the underlying disorder persists, the clinical syndrome of Delirium gradually shifts to some other, more stable, organic brain syndrome, or may cause death.

**Complications.** Injuries may be sustained from falling out of bed or attempting to flee frightening hallucinations. Agitation may interfere with proper medical management. If inadequately or belatedly treated, Delirium may lead to a Dementia or, more rarely, an Amnestic Syndrome or Organic Personality Syndrome.

**Etiological factors.** The causes of Delirium usually lie outside the nervous system and include: systemic infections; metabolic disorders such as hypoxia, hypercarbia, hypoglycemia, ionic imbalances, hepatic or renal disease, or thiamine deficiency; postoperative states; and substance intoxication and withdrawal. Delirium also occurs in hypertensive encephalopathy, following seizures, and on regaining consciousness after head trauma. Certain focal lesions of the right parietal lobe and inferomedial surface of the occipital lobe may present as Delirium.

**Predisposing factors.** The immature or senescent brain is more susceptible to the development of Delirium. Preexisting brain damage or a previous history of Delirium appears to increase the chances of an individual's developing this syndrome.

**Differential diagnosis.** Schizophrenia, Schizophreniform Disorder, and other psychotic disorders may also be marked by hallucinations, delusions, and disordered thinking and speech. In Delirium, however, these symptoms are extremely random and haphazard, without evidence of systematization. The course fluctuates, and there is evidence of a clouded state of consciousness with global cognitive impairment. Finally, in Delirium there is often a generalized slowing of background activity in the electroencephalogram, and the syndrome's cause is obviously organic.

Whereas **Dementia** involves a global cognitive deficit occurring in a normal state of consciousness, Delirium is basically a clouded state of consciousness.

Often, however, the two syndromes coexist in the same individual, and it may be difficult to decide how much of the clinical picture to ascribe to one syndrome and how much to the other. One cannot diagnose Dementia in the presence of significant Delirium, because the symptoms of Delirium interfere with the proper assessment of Dementia. Only a definite history of pre-existing Dementia allows one to decide that an individual with Delirium also has Dementia. When there is uncertainty as to whether the symptoms in a given individual are basically those of Delirium or Dementia, it is best to make a provisional diagnosis of Delirium. This should lead to a more active therapeutic approach, and with time the proper diagnosis will become apparent.

**Factitious Disorder with Psychological Symptoms** simulating an Organic Brain Syndrome might, under rare circumstances, present a problem in the differential diagnosis of Delirium. The individual with Factitious Disorder shows inconsistencies in tests of mental status. A normal electroencephalogram also helps to exclude Delirium.

### **Diagnostic criteria for Delirium**

A. Clouding of consciousness (reduced clarity of awareness of the environment), with reduced capacity to shift, focus, and sustain attention to environmental stimuli.

B. At least two of the following:

- (1) perceptual disturbance: misinterpretations, illusions, or hallucinations
- (2) speech that is at times incoherent
- (3) disturbance of sleep-wakefulness cycle, with insomnia or daytime drowsiness
- (4) increased or decreased psychomotor activity

C. Disorientation and memory impairment (if testable).

D. Clinical features that develop over a short period of time (usually hours to days) and tend to fluctuate over the course of a day.

E. Evidence, from the history, physical examination, or laboratory tests, of a specific organic factor judged to be etiologically related to the disturbance.

### **Dementia**

The essential feature is a loss of intellectual abilities of sufficient severity to interfere with social or occupational functioning. The deficit is multifaceted and involves memory, judgment, abstract thought, and a variety of other higher cortical functions. Changes in personality and behavior also occur. The diagnosis is not made if these features are due to clouding of consciousness, as in Delirium. However, Delirium and Dementia may coexist.

As with all Organic Brain Syndromes, an underlying causative organic factor is always assumed. In certain clinical states, e.g., Primary Degenerative Dementia, however, it may be impossible to show a *specific* organic factor as the definitive cause of the disturbance. These conditions may nevertheless be diagnosed as Dementia if (a) the impairment is a multifaceted loss of intellectual ability, (b) there is no evidence for a diagnosis other than an Organic Mental Disorder, and (c) a diligent search has failed to reveal a specific organic etiologic factor.

In the past, the term Dementia often implied a progressive or irreversible course. The definition of Dementia in this manual, however, is based on clinical symptoms alone, and carries no connotation as to prognosis. Dementia may be progressive, static, or remitting. The reversibility of a Dementia is a function of the underlying pathology and of the availability and timely application of effective treatment.

Memory impairment is usually the most prominent symptom. In mild Dementia there is forgetfulness in daily life and a need to have statements repeated several times to facilitate remembering. On examination there may only be a certain hesitancy in response to questions. In more severe memory impairment, the individual may forget names, telephone numbers, directions, conversations, and events of the day. He or she may leave tasks unfinished because of forgetting to return after an interruption. This may cause a person to leave the water running in the sink or to neglect turning off the stove. In advanced forms of Dementia, memory impairment is often so severe that the person forgets the names of close relatives, his or her own occupation, schooling, birthday, or, occasionally, even his or her own name. Memory disturbance may be formally documented by demonstrating difficulty in learning new information (short-term memory deficit) and in recalling material that was known in the past (long-term memory deficit). The former is tested by asking the individual to memorize the names of several unrelated objects, or a brief sentence, and then to repeat them after a few minutes of distraction; the latter is tested by asking about events that happened in the past.

Impairment of abstract thinking takes many forms. The individual has trouble coping with novel tasks, especially if pressed for time. He or she may try to avoid situations and tasks that require the processing of new and complex information. This deficit is sometimes formally assessed by asking the individual to interpret proverbs or to perform such tasks as finding similarities or differences between related words. The individual with Dementia interprets proverbs concretely and has difficulty in finding similarities or differences.

Impaired judgment and impulse control are also commonly observed. Coarse language, inappropriate jokes, neglect of personal appearance and hygiene, and a general disregard for the conventional rules of social conduct are evidence of bad judgment and poor impulse control. A previously cautious businesswoman may embark on a reckless business venture. An elderly spinster may make sexual advances to strangers. A retiree may shoplift without considering the consequences. Marked impairment of judgment and impulse control is particularly characteristic of certain Dementias that affect primarily the frontal lobes.

Dementia also involves a variety of disturbances of higher cortical function.

Although language is unaffected by some neurological disorders that cause Dementia, in others it is abnormal. It may appear vague, stereotyped, and imprecise, with long circumlocutory phrases; or there may be specific signs of aphasia, such as difficulty naming objects. In severe forms of Dementia, the individual may be virtually mute. So-called "constructional ability" is nearly always disturbed, and can be demonstrated by having the individual copy three-dimensional figures, assemble blocks, or arrange sticks in specific designs. Agnosias (failure to recognize or identify objects despite intact sensory function) and apraxias (inability to carry out motor activities despite intact comprehension and motor function) may also be present.

Personality change is almost invariably present in Dementia, and may involve either an alteration or an accentuation of premorbid traits. A common pattern is for a normally active individual to become increasingly apathetic and withdrawn. The range of social involvement narrows. The personality loses its sparkle, and the individual is described by others as "not himself (or herself)." Another pattern of change is for a previously neat and meticulous person to become slovenly and unconcerned about appearances. On the other hand, some individuals display an accentuation of preexisting compulsive, histrionic, impulsive, or paranoid traits. Irritability and cantankerousness are also common features of Dementia.

**Associated features.** When Dementia is mild and the individual has some grasp of his or her deteriorating faculties, he or she may react with marked anxiety or depression. Attempts to conceal or compensate for subjectively perceived intellectual deficits are very common. This may result in excessive orderliness, social withdrawal, or a tendency to relate events in excessive detail so as to avoid exposure of gaps in memory. Paranoid ideation may occasionally be quite marked and result in false accusations and verbal or physical attacks. The habitually jealous individual who develops a Dementia may develop the delusion of marital infidelity and actually assault his or her spouse.

Individuals with Dementia are especially vulnerable to physical and psychosocial stressors. For example, bereavement or retirement may considerably aggravate intellectual deficits.

**Age at onset.** Dementia is found predominantly in the elderly, although certain specific etiologic factors (see below) may cause Dementia at any age. The diagnosis of Dementia may be made at any time after the intellectual quotient is fairly stable (usually by age 3 or 4). Thus, if a child at age 4 developed a chronic neurological disorder that interfered with previously acquired functions so as to significantly lower intellectual and adaptive functioning, he or she would be considered to have both Dementia and Mental Retardation. In such a case both diagnoses should be made, Mental Retardation being listed first because of its greater relevance to management.

**Course.** The mode of onset and subsequent course of Dementia depend on the underlying etiology. When Dementia is a result of some clearly defined episode of neurological disease, such as cerebral hypoxia or encephalitis, or of head

trauma, it may begin quite suddenly, but then remain relatively stationary for a long period of time. Primary Degenerative Dementia, on the other hand, is usually insidious in onset and slowly, but relentlessly, progresses to death over a period of several years. Dementia resulting from brain tumors, subdural hematomas, and metabolic factors may also have a gradual onset. When the underlying disorder can be treated, as in hypothyroidism, subdural hematoma, normal-pressure hydrocephalus, and tertiary neurosyphilis, Dementia may be arrested or even reversed. However, the more widespread the structural damage to the brain, the less likely is clinical improvement.

**Impairment.** By definition Dementia is diagnosed only when the loss of intellectual function is sufficiently severe to interfere with social or occupational functioning, although the degree of impairment may vary. In advanced Dementia the individual becomes totally oblivious of his or her surroundings and requires constant care.

**Complications.** Individuals with Dementia may wander and become lost. They may, occasionally, do harm to themselves or others. Delirium frequently is a complication of Dementia. Individuals with severe Dementia are susceptible to infectious diseases, which often prove fatal.

**Etiological factors.** Primary Degenerative Dementia of the Alzheimer type is the most common Dementia. Other causes include: central nervous system infections (including tertiary neurosyphilis, tuberculous and fungal meningitis, viral encephalitis, and Jakob-Creutzfeldt disease); brain trauma (especially chronic subdural hematoma); toxic-metabolic disturbances (such as pernicious anemia, folic-acid deficiency, hypothyroidism, bromide intoxication); vascular disease (Multi-infarct Dementia); normal-pressure hydrocephalus; neurological diseases such as Huntington's chorea, multiple sclerosis, and Parkinson's disease; and postanoxic or posthypoglycemic states.

**Differential diagnosis.** The normal process of aging has been associated in a number of studies with a variety of different changes in intellectual function. The nature of these changes and whether they should be considered true decrements of function, however, remain controversial. The diagnosis of Dementia is warranted only if intellectual deterioration is of sufficient severity to interfere with social or occupational functioning. Dementia is not synonymous with aging.

In **Delirium** there is also impairment of intellectual abilities, but it occurs in the context of clouding of consciousness; in Dementia the state of consciousness is normal. The clinical course of these two syndromes also differs. In Delirium symptoms typically fluctuate, whereas in Dementia they are relatively stable. An Organic Mental Disorder persisting in unchanged form for more than a few months suggests Dementia rather than Delirium. (See also Delirium, Differential diagnosis, p. 106.)

**Schizophrenia**, especially when chronic, may be associated with some degree of intellectual deterioration. The absence of identifiable brain pathology helps rule out the additional diagnosis of Dementia.

Individuals with a **major depressive episode** may complain of memory impairment, difficulty in thinking and concentrating, and an overall reduction in intellectual abilities. They may also perform poorly on mental-status examination and neuropsychological testing. These features may suggest the possible diagnosis of Dementia, and this phenomenon is sometimes known as "pseudodementia." Depression, however, is primarily a disturbance of mood. Any cognitive deficits observed may be viewed as secondary to the disturbed affect. If sufficiently motivated to perform, individuals with depression usually demonstrate intact cognitive function. Dementia, on the other hand, is basically a disorder of intellectual function. Abnormalities of mood are less frequent and, when present, less pervasive than in depression. The clinical history also helps to differentiate between the two. In depressive pseudodementia, the onset can frequently be dated with some precision, and symptoms progress more rapidly than in true Dementia. In addition, there may be a history of previous mental illness. On formal mental-status testing there may be considerable variability in performance as opposed to the more consistently poor performance of individuals with Dementia. In the absence of evidence of a specific organic etiologic factor, if the features suggesting major depressive episode are at least as prominent as those suggesting Dementia, it is best to diagnose major depressive episode and to assume that the features suggesting Dementia are secondary to the depression. A therapeutic trial with an antidepressant drug or electroconvulsive therapy (ECT) (if not contraindicated) may clarify the diagnosis in that if the disorder is actually a major depressive episode, cognitive impairment may resolve as the mood improves.

**Factitious Disorder with Psychological Symptoms** may mimic Dementia, but rarely.

#### **Diagnostic criteria for Dementia**

A. A loss of intellectual abilities of sufficient severity to interfere with social or occupational functioning.

B. Memory impairment.

C. At least one of the following:

(1) impairment of abstract thinking, as manifested by concrete interpretation of proverbs, inability to find similarities and differences between related words, difficulty in defining words and concepts, and other similar tasks

(2) impaired judgment

(3) other disturbances of higher cortical function, such as aphasia (disorder of language due to brain dysfunction), apraxia (inability to carry out motor activities despite intact comprehension and motor function), agnosia (failure to recognize or identify objects despite intact sensory function), "constructional difficulty" (e.g., inability to